

A 0-TO-1 INTRODUCTION TO QUANTUM COMPUTING · WITH QUBIS

Quantum from Zero to One

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Photo: Qolour



THE OPPORTUNITY

Over \$12 billion invested in 2025 alone

A field that's touted to change the world as we know it:



New medicines

Simulating molecules
regular computers can't



New energy sources

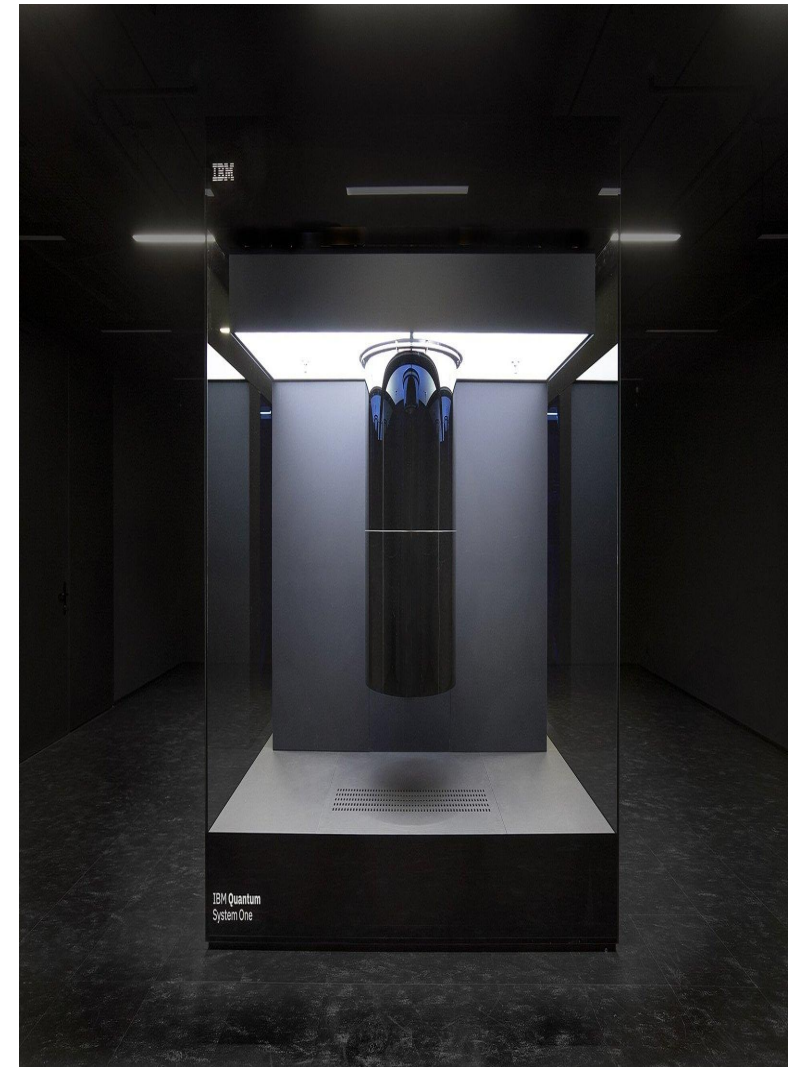
Designing better materials
and catalysts



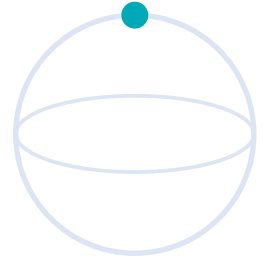
Breaking encryption

Today's codes rest on hard
math problems

And more.

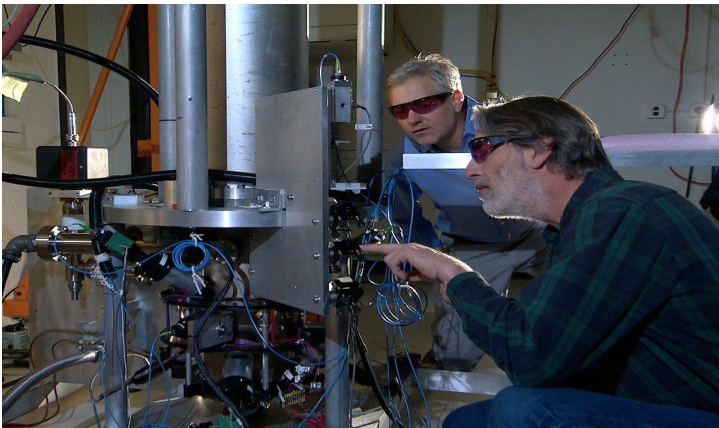


IBM Q System One at Fraunhofer · IBM Research, CC BY 2.0 (Wikimedia Commons)



A century of quantum technology

Quantum physics has transformed our understanding of reality, and made its way into technology that improved our quality of life:



NIST-F2 cesium fountain clock · NIST, public domain

Atomic clocks

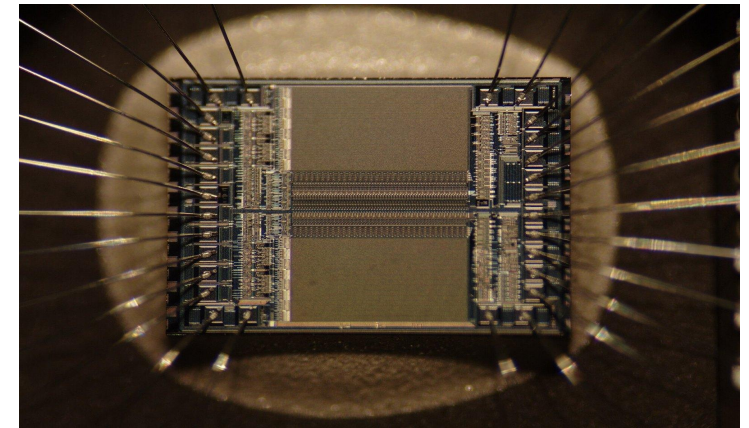
Precise time — and the GPS you use for navigation



MRI suite · U.S. Marine Corps, public domain

MRIs

Highly detailed scans of the inside of your body



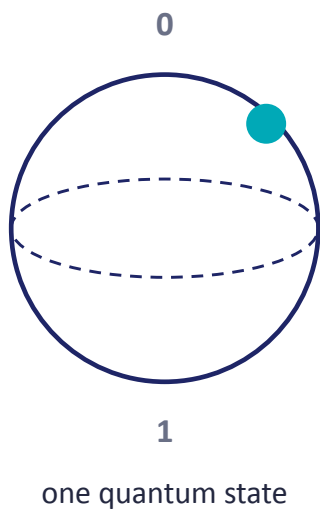
Integrated circuit macro · Zephyris, CC BY-SA 3.0

Transistors

The backbone of all modern electronics

A new wave: quantum 2.0

Quantum 2.0 is about isolating, manipulating, and measuring individual quantum states — like the spin of a single electron.



isolate → manipulate → measure



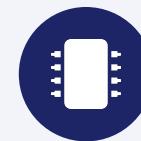
Sensors

Higher resolution



Networks

More robust



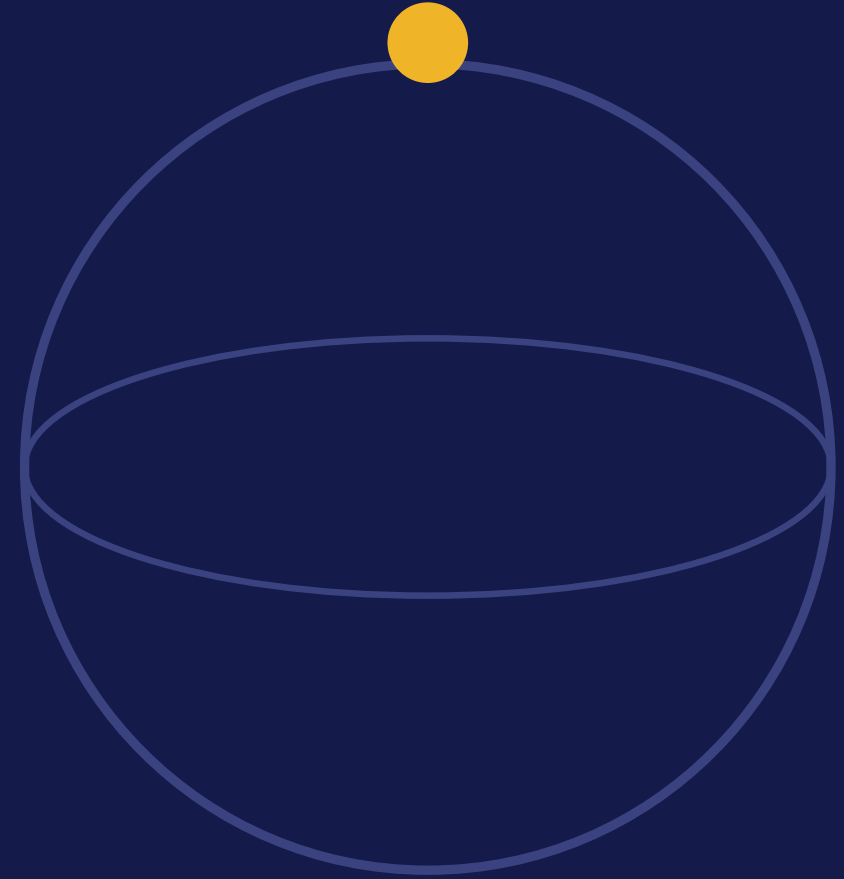
Computers

Quantum

THE QUESTION

What makes quantum computing special?

How is it different from regular computers?



Regular computers are based on bits

Bits are the memory element of a computer. It's how computers store information.

The gist of what a computer does is manipulate this information in a useful way to us.

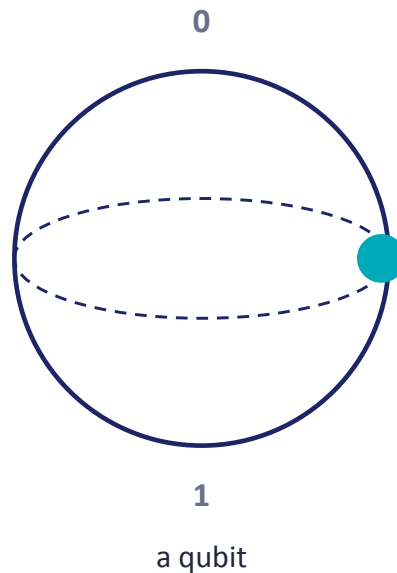
A BIT IS EITHER 0 OR 1



Quantum computers: the same idea

They have some memory element that stores information.

The gist of what quantum computers do is just manipulate the information stored in these memory elements.



The difference: the memory element for a quantum computer is a qubit.

QUBIS

These are model qubits.

Qubi by Qolour — a handheld qubit you can hold, flip, and read.

Photo: Qolour

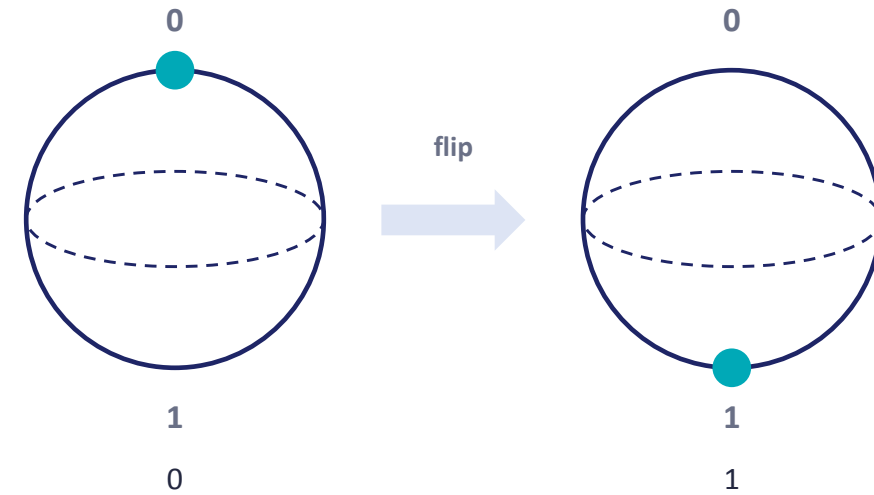


SIMILARITY 1

Bits can be 0 or 1. So can qubits.

On the Qubi:

- 1 The top we call 0
- 2 The bottom we call 1
- 3 Just like with bits, we can flip from 0 to 1 and 1 to 0



SIMILARITY 2

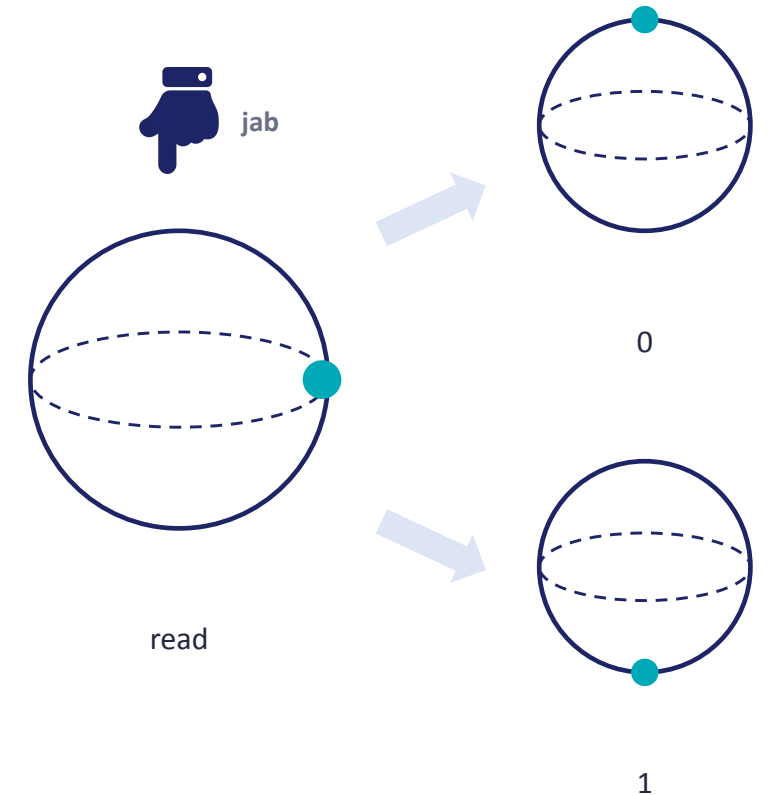
Bits and qubits can be read

If information is stored in a bit or qubit, the computer can read it and take it somewhere else.

1 To read the Qubi, jab it straight downwards

2 If it goes to the top, that's a 0

3 If it goes to the bottom, that's a 1

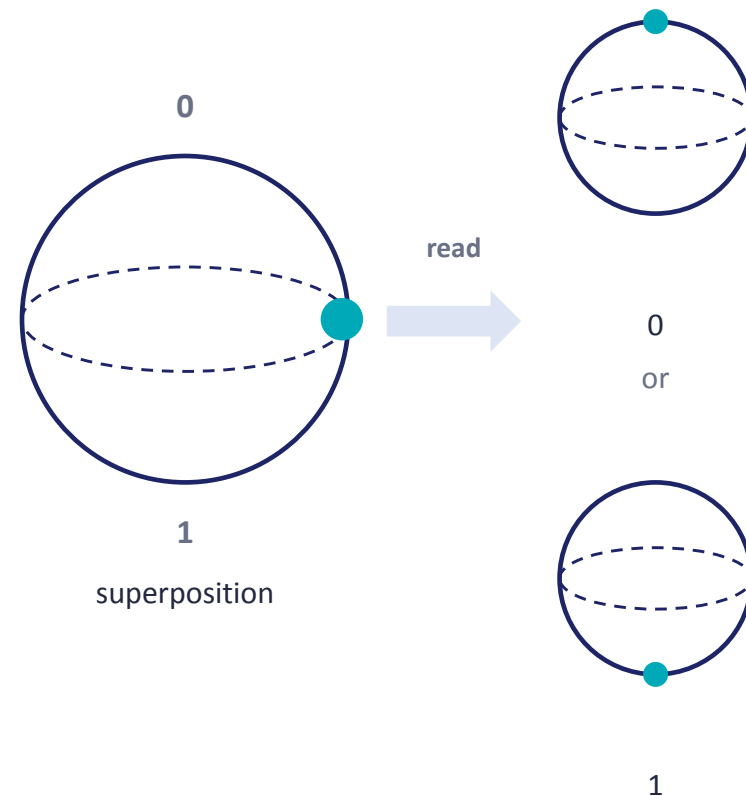


A qubit can also be between 0 and 1

A bit is either 0 or 1. A qubit can also be between 0 and 1 — a superposition.

Watch what happens when I read a qubit in superposition: it goes back to 0 or 1.

A qubit can be in superposition until you read it. At the end of the day, the answer you get is still 0 or 1.

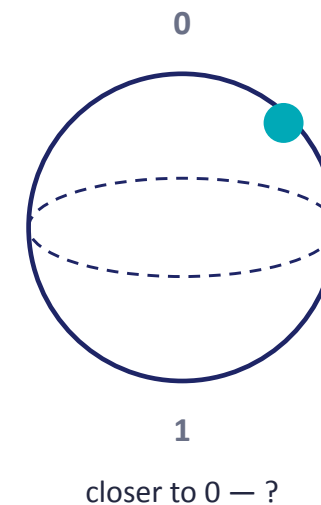
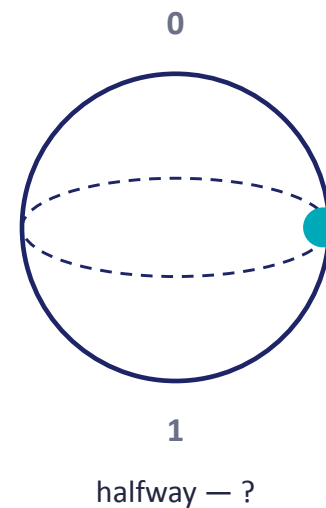


What are the chances?

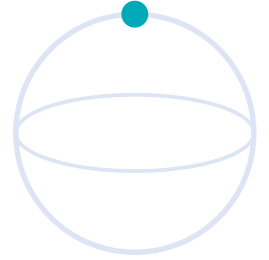
Two questions:

1 Exactly halfway between 0 and 1 — what are the chances it goes to 0 or 1 when I read it?

2 Closer to 0 — now what are the chances it goes to 0?



Two qubits



Put two qubits in close proximity — by bumping them — and they can become quantumly entangled.

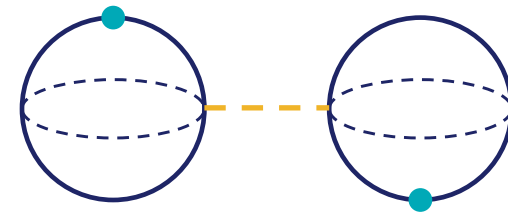
- 1 Bump the two Qubis together
- 2 Read one qubit
- 3 The other qubit instantly collapses to the opposite side
- 4 Move one away — even to the other side of the universe — and do it again

Many types of entanglement

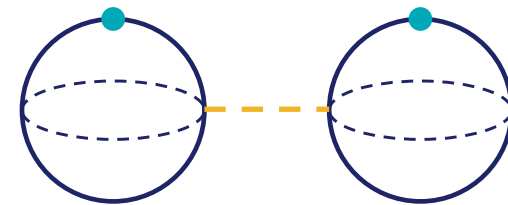
For example:

- 1 Bump the Qubis together, then read them → opposite results
- 2 Bump the Qubis together and flip one, then read → the same result
- 3 Entanglement can be made with any number of qubits, not just 2

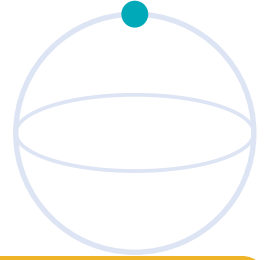
SINGLET — opposite results



PHI+ (Φ^+) — same results



What makes quantum computing different?



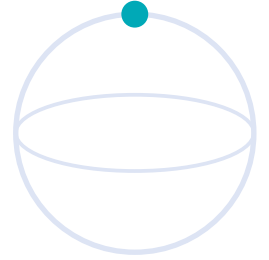
STILL THE SAME

- A memory element
- Stores 0s and 1s
- Reads them out



COMPLETELY DIFFERENT

- Superposition — a qubit can be between 0 and 1
- Entanglement — qubits can be entangled so that one depends on the other



Problems regular computers can't solve

Superposition and entanglement might look like small tricks. Together, they let quantum computers take on problems that are extremely difficult for regular computers:



Simulating chemical reactions

Molecules are quantum systems. A quantum computer can model them directly — new medicines, new materials, new sources of energy.



Breaking encryption

Today's encryption relies on math problems that take regular computers longer than the age of the universe. Quantum algorithms change that.

QUESTIONS

What questions do you have?

